



No. of Printed Pages: 1

Seat No : _____

[92N]

SARDAR PATEL UNIVERSITY
BCA Semester-2 External Examination – April-2024 [NEP]

US02IDBCA05 || Digital Electronics
Time : 02.00 p.m. – 03.30 p.m.

Total : 25 Marks

Date : 13-04 -2024

Q-1 Select the appropriate option for the following questions:

[05]

- 1) The _____ gate has two or more input signals. All inputs must be high to get a high output.
A. AND
B. OR
C. NAND
D. NOR
- 2) The NOR gate has two or more input signals. If all inputs are _____, the output is high.
A. High
B. Low
C. Both A and B
D. None of these
- 3) In half adder AND gate's output is _____.
A. Carry
B. Sum
C. Remainder
D. None of these
- 4) What is the one disadvantage of an RS Flip-Flop has _____.
A. no Enable input
B. a RACE condition
C. no clock input
D. only single output
- 5) The length of a register is called _____.
A. Word limit
B. Word size
C. Register limit
D. Register size

Q-2 What is logic gate? Draw electronic symbol and explain AND, OR, NOT and XOR gates with truth table. [10]

OR

Q-2 Explain Decoder with circuit and truth table. [10]

Q-3 Explain RS Flip-Flop with diagram. [10]

OR

Q-3 Explain Left-Shift and Right-Shift register with diagram. [10]

— X —